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DEEMED-TO-BE UNIVERSITY

आविष्कार
FET HACKATHON

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CONNECT | INNOVATE | IMPACT

Different Indian Languages Same Intent **Stronger AI.**

Same Code More Perspectives Bigger Possibilities.

EL Rata Alada

</> CROSS-LINGUAL CODE-SWITCHING ROBUSTNESS TESTER </>

यह code सही है ?

এইটো কাজ কৰেনে ?

இந்த code-ல் ஏதேனும் error ?

ಈ function ಫಿಕ್ಸ್ ಮಾಡಿದೆಯಾ ?

Different Indian Languages Same Questions Same Possibilities

More Indian Voices More Inclusive AI.



Team Members

Tanmay
Team Lead & Development
Builds the app, runs the live demo, answers technical questions, writes Manglish prompts.

Ritobroto
Testing & Evaluation
Runs the AI models, writes the grading test cases, results tables, Benglish prompts.

Aaditya
Dataset & Design
Prepares dataset, handles coding tasks and Hinglish prompts, deck and dashboard design.

Aaliya
Research & Presentation
Leads the pitch, problem-story research, judge Q&A, Odia prompts.

Problem Statement

314.9M

Indians are bilingual

Census of India, 2011

Many think and type in mixed languages, including when they ask AI to write code.

“yeh function crash kyu ho raha hai?”

THE PROBLEM

AI coding assistants are tested in clean English, but Indian developers ask in Hinglish, Benglish, Kanglish, Tanglish and Odia-English.

Tested in English, used in Hinglish

Code-generation benchmarks like HumanEval-XL use single-language prompts, and none of its 23 languages is Indian.

Nobody measures the gap

No one knows how much worse the generated code gets when the prompt is mixed, so students and developers get broken code without knowing why.

Solution

MixCheck gives an AI the same coding task in English and in mixed Indian languages, runs the code it writes, and shows how much the pass rate drops.

STEP 1

Paired prompts

25 coding tasks, each written in English and in Hinglish, Benglish, Kanglish, Tanglish and Odia-English: 10 versions of every task.

STEP 2

Run and grade

Every version goes to 3 AI models (Gemini, GPT-OSS 120B and 20B). Their code is run against the same test cases.

STEP 3

See the drop

A Streamlit dashboard compares English with each mix, opens every failure, and runs a live face-off between models.

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Tech Stack

Frontend / Dashboard

Streamlit Plotly

- Web app & interactive UI
- Input prompts & select language
- Visualize results with charts

Backend / Core

Python

- Everything: prompts, running code, scoring & core logic
- Modular and easy to extend

Data Handling

pandas

- Prompt sets (one CSV per language)
- Store results tables
- Data processing & analysis

TECH STACK

AI Models to Test

Gemini API Groq

- Coding assistants we test
- Both have free tiers
- Send prompts & get code output

AI Models Tested

Gemini · GPT-OSS 120B · GPT-OSS 20B

- Same prompt to every model
- Code run in its own process

Testing / Evaluation

Python unittest

- Runs the AI's code against test cases
- Pass or fail
- Same test cases for English & mixed-language prompts

Collaboration & Development

GitHub
- Version control & collaboration
- Shared code

Claude
- AI pair programmer
- Helps with development, debugging and improvements

Workflow

01 User Input

Same coding task in English and four mixed-language prompts (Hinglish, Benglish, Tanglish, Odia-English).

- yeh function crash kyu ho raha hai?
- ei list ta sort kore dao
- indha code-la enna error?
- ei function ta fix karidia

02 Send to AI Models

Send each version of the prompt to the AI models (Gemini API and Groq).

Gemini API Groq

03 Run Generated Code

Execute the code produced by the models against the same test cases using Python unittest.

Python unittest

04 Compare Results

Compare how often the code passes (English vs mixed-language).

- Pass rate comparison
- Error analysis
- Visual charts (Plotly)
- Language-wise insights

05 Show on Dashboard

Display results, charts and key insights in the Streamlit dashboard.



OPEN-SOURCE & STUDENT-FRIENDLY
Free tools, easy to run on laptops

BUILT FOR INDIAN LANGUAGES
Hinglish, Benglish, Tanglish, Odia-English

SCALABLE & EXTENSIBLE
Add more Indian languages and models later

Innovation

Paired prompts

Same task, only the language changes, so any drop in pass rate comes from the mixing.

Two scores, both automatic

Test cases decide if the code works. A second check asks: did the AI follow the instructions?

Indian languages first

Covers Kanglish and Odia-English, low-resource mixes, alongside Hinglish, Benglish and Tanglish.

Fills a real gap

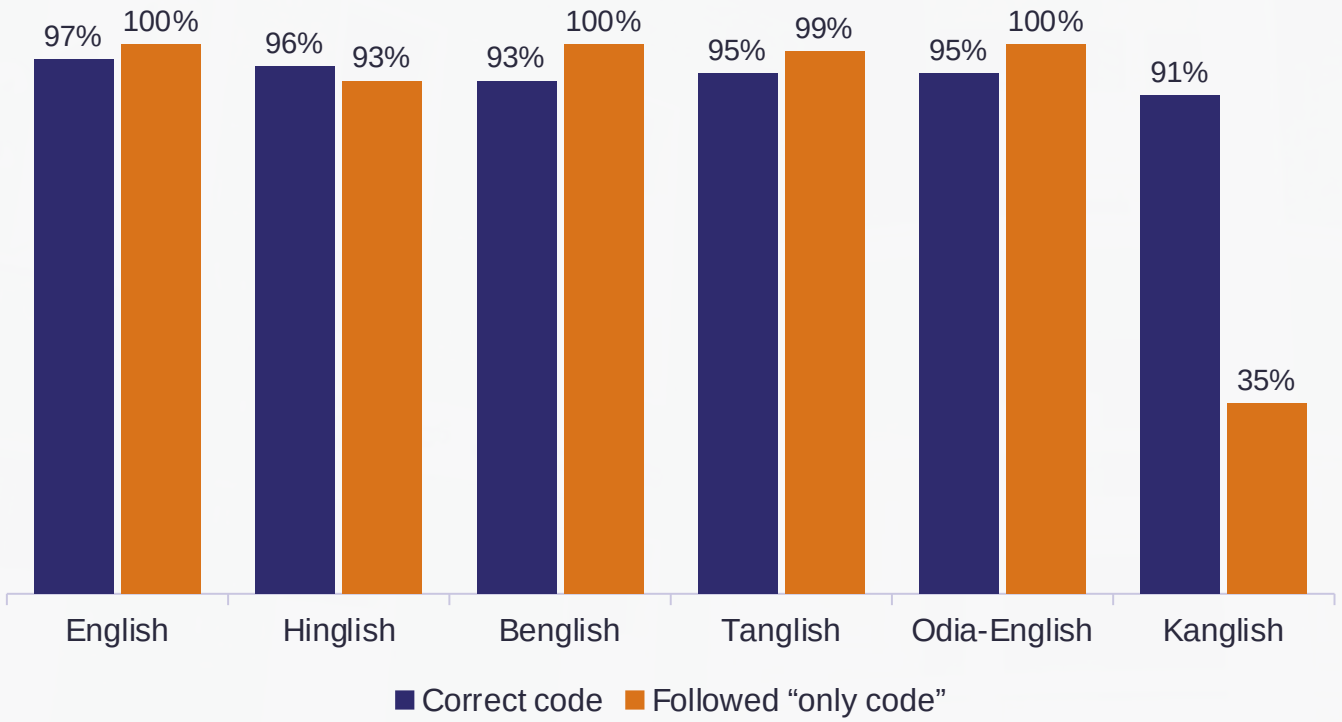
HumanEval-XL covers 23 languages, none Indian and none mixed. LinCE and GLUECoS don't test code generation.

Every prompt is written or checked by a native speaker.

Results

Same casual style, only the language changes

Share of answers, across 3 AI models and 25 tasks



35%

of Kanglish answers replied with only the code, as asked. Casual English: 100%.

9 vs 0

wrong function names in 600 mixed prompts vs 150 English ones.

750

answers graded by running the code: 25 tasks × 10 versions × 3 models.

One run per prompt at temperature 0. The small gaps in correct code are signals; the Kanglish instruction gap is large.

Where the AI Slips

FAILURE 1

Renames your function

9 of 600 mixed prompts, 0 of 150 English. The name was in every prompt.

```
bhai indian_format naam er
python fn bana…
def bhai_indian_format(n):
```

FAILURE 2

Ignores “only code”

Kanglish: GPT-OSS 120B obeyed 0 of 25 times. 10 answers drifted into Kannada script.

```
…bari code  (only code)
→ ~2,000 characters
→ 2 code blocks + notes
```

FAILURE 3

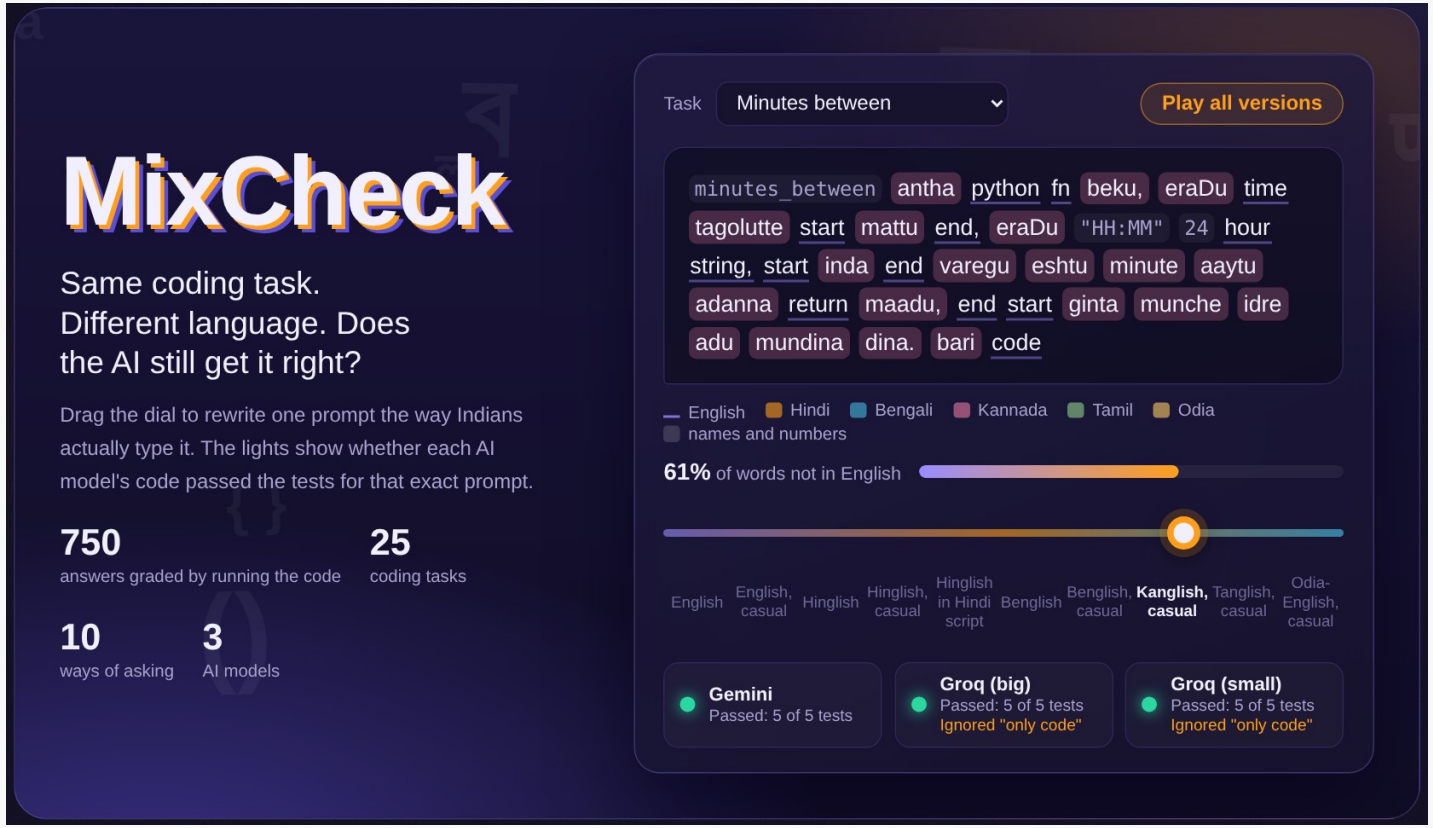
Skips a detail

The “ignore upper and lower case” rule was dropped in casual Benglish and Odia.

```
is_palindrome("Racecar")
expected  True
got       False
```

The code mostly still works. The instructions break: Kanglish 35% vs English 100%.

Live Demo



The MixCheck dashboard, built with Streamlit and Plotly

Turn the Mix Dial

One prompt rewritten from English to Kanglish. Lights show which AI still passed.

Open any failure

Click a cell to read the AI's exact answer and the test it failed.

Face-off, live

Send a mixed prompt to two models right now and grade both answers.

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Impact

OF CROSS-LINGUAL CODE-SWITCHING ROBUSTNESS TESTER

01 More Inclusive AI Systems

Helps ensure AI coding tools work effectively for mutiilingual and code-switching users, including those who mix languages like English, Hindi, Tamil, and others.



02 Improved Model Reliability

Identifies failure cases and weaknesses in LLMs when handling code-switching, leading to more robust, reliable, and real-world ready AI coding assistants.



03 Better Developer Experience

Reduces frustration for developers who naturally mix languages, enabling smoother, more accurate code understanding and assistance.



04 Supports Real-World Use Cases

Reflects actual developer behavior in diverse settings such as students, startups, freelancers, and global teams where code-switching is common.



05 Bridging Language Barriers

Makes advanced AI coding tools accessible to non-English-speaking regions, promoting participation and innovation from a global developer community.



06 Encourages Fair & Ethical AI

Helps detect and reduce linguistic bias in AI models by evaluating performance across languages and code-switching scenarios, leading to more equitable and inclusive AI systems.



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Target Audience / Market

FOR CROSS-LINGUAL CODE-SWITCHING ROBUSTNESS TESTER

Bilingual Population in India

314.9 Million (26%)

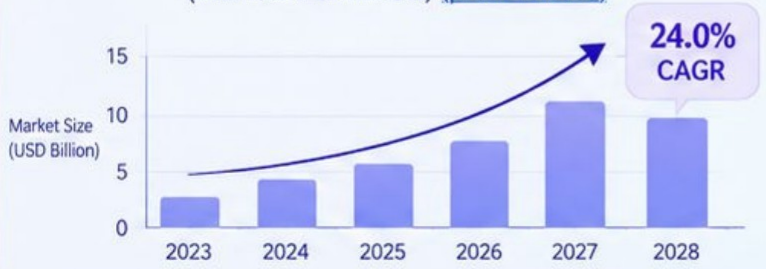
Indians are bilingual
(Census of India 2011)
(Multilingualism in India)



Growing Market for AI Code Tools

USD **4.3 Billion** in 2023 → USD **12.6 Billion** by 2028

24.0% CAGR
(MarketsandMarkets) (press release)



Focus on Indian Languages

We evaluate AI coding assistants on four Indian code-switched language pairs.

Hinglish
yeh function crash kyu ho raha hai?

Benglish
ei list ta sort kore dao

Tanglish
indha code-la enna error?

Odia-English
ei function ta fix karidia

Key User Segments (Target Audience)



Students & Learners
Use mixed language prompts while learning and building projects



Aspiring Developers
Practise and build with multilingual prompts



Educators & Researchers
Need robust evaluation for multilingual code understanding



Organizations & Enterprises
Build more inclusive AI tools for Indian and global users

Gap in Existing Benchmarks

23 Languages
tested in HumanEval-XL, but none are Indian and there are no mixed-language prompts.
(HumanEval-XL on GitHub)



Single-language prompts only
No Indian languages
No code-switched prompts

Geographic Relevance



Future Opportunity

More languages later:
Spanish, Chinese, Japanese, Korean and Arabic can be added in future work.

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Revenue Model

CROSS-LINGUAL CODE-SWITCHING ROBUSTNESS TESTER

Sustainable growth by serving individuals, teams and organizations building AI for a multilingual world.

Revenue Streams

1 Freemium (Individuals)

Free tier with basic testing and limited language combinations.

2 Pro (Developers & Small Teams)

Paid subscription with higher usage limits and advanced features.

3 Enterprise (Organizations)

Custom plans for companies, research labs and institutions with large-scale testing needs.

4 API Access

Usage-based pricing for integrating our robustness tester into existing ML pipelines.

Pricing Tiers (Illustrative)

Free
For learners & hobbyists

₹0

- Basic testing
- Limited languages (Indian pairs only)
- Standard reports

Get Started

Pro
For developers & small teams

₹799 / month

- More language pairs (Hinglish, Benglish, Tanglish, Odia-English)
- Detailed reports
- Priority support

Subscribe

Enterprise
For organizations & research

Custom Pricing

- High-volume testing
- Custom language sets (Indian languages)
- Dedicated support
- On-prem/Cloud options

Contact Sales

API
For integration

Pay-as-you-use

- Per request pricing
- Scalable infrastructure
- Detailed usage analytics
- Integrate into existing pipelines

View API Docs

How Each Segment Generates Revenue



Individuals
(Learners / Students)

Freemium → Pro conversion



Developers
(Small Teams)

Monthly / Annual Subscriptions



Organizations
(Enterprises, Research Labs)

Custom Licensing & Support Plans



API Users
(Platforms / Integrators)

Usage-based (API Calls)

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Future Improvements

CROSS-LINGUAL CODE-SWITCHING ROBUSTNESS TESTER

01 Expand Language Coverage



English | Hindi | Tamil | Bengali | Spanish | Arabic | ...

Support more regional, low-resource, and mixed-language combinations beyond current coverage.

02 Smarter Code-Switch Detection

```
# calculate total
अमर value > 10:
    return resultado
```

English Hindi Spanish

Automatically identify where language switching occurs in code, comments, prompts, and natural language.

03 Deeper Robustness Testing



- Syntax
- Logic
- Security
- Runtime
- Performance

Go beyond syntax to test logic, security, runtime behaviour, and performance under mixed-language inputs.

04 Real-Time Testing

```
def process_data(df):
  # डेटा को साफ करें
```

Testing...

- Robust
- No critical issues
- Suggestions (2)

Provide instant robustness feedback while developers interact with AI coding assistants.

05 Advanced Analytics



- Error Rate
- Failure Cases
- Language Pairs
- Severity

Top Failure Scenarios

1.	Hindi + English	32%
2.	Spanish + English	24%
3.	Tamil + English	18%

Track recurring failure patterns and show which languages and switching scenarios cause problems.

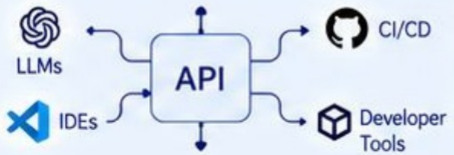
06 Adaptive AI Testing



- New scenarios
- Previous results
- Harder test cases

Use previous test results to continuously generate new and more challenging code-switching scenarios.

07 Easy API Integration



Allow the tester to plug directly into existing LLMs, IDEs, CI/CD pipelines, and developer tools.

08 More Inclusive AI



Improve support for underepresented languages and make AI coding tools accessible to a wider developer community.

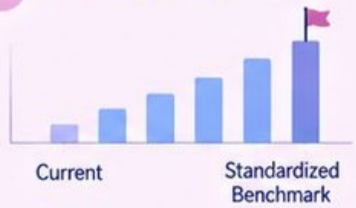
09 Continuous Safety Evaluation



- Bias & toxicity
- Prompt injection
- Unexpected behaviour
- Linguistic edge cases

Regularly test models against new linguistic edge cases, vulnerabilities, and unexpected behaviour.

10 Global Benchmark



Build a standardized benchmark for measuring cross-lingual robustness of AI coding systems.

Vision

From understanding one language to understanding how developers actually communicate — mixed, multilingual, and real-world.

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Thank You

Same Code Different Languages Stronger AI.

Test Understand Improve

```
def analyze(text):
    # mixed language input
    if भाषा == "हिंदी":
        return "robust"
    else:
        return "test further"
```

Multilingual Developers Stronger Tools Brighter Possibilities

अ यह code सही है?

অ এটা ঠিক কাজ করছে?

Different Languages Same Intent

க இந்த code சரியாக இயங்குகிறதா?

ଏ ଏହା code ଠିକ୍ ଅଛି କି?

ଓଡ଼ିଆ

हिन्दी

தமிழ்

বাংলা

Questions?

SAME LANGUAGES. NEW PERSPECTIVES. STRONGER AI.

BUILDING A MORE INCLUSIVE AI TOMORROW

Team

Tanmay
TEAM LEAD & DEVELOPMENT

- Builds the app with Claude
- Runs the live demo
- Answers technical questions
- Writes Tanglish prompts

Aaditya
DATASET & DESIGN

- Prepares coding tasks
- Writes Hinglish prompts
- Deck and dashboard design
- Dataset curation & structuring

Aaliya
RESEARCH & PRESENTATION

- Leads the pitch
- Problem-story research
- Judge Q&A
- Writes Odia prompts

Ritobroto
TESTING & EVALUATION

- Runs the AI models
- Writes the grading test cases
- Prepares results tables
- Writes Benglish prompts
- Has the most powerful laptop